DAY 4: INTRODUCTION TO OSPF

Cisco Packet Tracer Practical Report

Objective

To configure Open Shortest Path First (OSPF) on three routers in a triangle topology, verify IP connectivity, observe OSPF neighbor adjacency, and examine how interface cost affects route selection.

Theory: How OSPF Works

OSPF (Open Shortest Path First) is a link-state interior gateway routing protocol. Routers running OSPF exchange link-state information and build a link-state database. They use the Shortest Path First (SPF) algorithm to calculate the lowest-cost path to each destination. OSPF metric is called cost; lower total cost is preferred.

Important Terms

- **Area:** A logical grouping of OSPF routers and links. Area 0 is the backbone area. This lab uses Area 0.
- **LSA (Link-State Advertisement):** Information OSPF routers share about their links and network topology.
- **Cost:** A metric associated with an interface. OSPF adds interface costs along a path and normally selects the lowest total cost.
- **Neighbor adjacency:** A relationship formed between OSPF routers so they can exchange link-state information. On broadcast Ethernet networks, routers may show FULL/DR or FULL/BDR; point-to-point links commonly show FULL/-.

Topology and IP Addressing

Create three routers (R1, R2, R3) connected in a triangle. Use Ethernet or serial links. The example below uses three /30 point-to-point subnets:

Link	Router interface	IP address	Subnet mask
R1–R2	R1 G0/0	10.0.12.1	255.255.255.252
R1–R2	R2 G0/0	10.0.12.2	255.255.255.252
R2–R3	R2 G0/1	10.0.23.1	255.255.255.252
R2–R3	R3 G0/0	10.0.23.2	255.255.255.252
R3–R1	R3 G0/1	10.0.13.2	255.255.255.252
R3–R1	R1 G0/1	10.0.13.1	255.255.255.252

Interface names vary by router model. If your router uses FastEthernet or serial interfaces, replace the interface names in the commands with the actual names shown in Packet Tracer.

Procedure

1. Open Cisco Packet Tracer and place three routers in the workspace.
2. Connect R1 to R2, R2 to R3, and R3 to R1 using Ethernet or serial links.
3. Configure the IP address and subnet mask on every connected interface. Enable each interface with the no shutdown command.
4. If using serial DCE links, set a clock rate on the DCE end if required by the router/link configuration.

5. Ping the directly connected neighbor IP addresses to verify basic connectivity before enabling OSPF.
6. Configure OSPF process 1 on each router and advertise the connected /30 networks in Area 0.
7. Wait for OSPF neighbors to form, then use show ip ospf neighbor and show ip route ospf to verify operation.
8. Change the OSPF cost on one link and check whether the preferred route changes.

Router Configuration Commands

R1

```
enable
configure terminal
hostname R1
interface gigabitEthernet0/0
 ip address 10.0.12.1 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 10.0.13.1 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 1.1.1.1
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.13.0 0.0.0.3 area 0
end
write memory
```

R2

```
enable
configure terminal
hostname R2
interface gigabitEthernet0/0
 ip address 10.0.12.2 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 10.0.23.1 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 2.2.2.2
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.23.0 0.0.0.3 area 0
end
write memory
```

R3

```
enable
configure terminal
hostname R3
interface gigabitEthernet0/0
 ip address 10.0.23.2 255.255.255.252
 no shutdown
exit
interface gigabitEthernet0/1
 ip address 10.0.13.2 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 3.3.3.3
 network 10.0.23.0 0.0.0.3 area 0
 network 10.0.13.0 0.0.0.3 area 0
end
write memory
```

Verification Commands

- **show ip interface brief —** Check interface addresses and ensure interfaces are up/up.
- **ping <neighbor-IP> —** Test IP connectivity between directly connected routers.
- **show ip ospf neighbor —** Display OSPF neighbors and their adjacency states.
- **show ip route ospf —** Display routes learned through OSPF (marked with O).
- **show ip protocols —** Check OSPF process settings and advertised networks.
- **show ip ospf interface —** Inspect OSPF status and cost on interfaces.

Observing OSPF Adjacency

Run show ip ospf neighbor on all three routers. On point-to-point links, a fully established neighbor commonly appears as FULL/-; on broadcast Ethernet segments, the state may be FULL/DR or FULL/BDR. Each router should eventually see the other two routers as neighbors in this triangle. The exact display depends on the interface/network type.

If no neighbor appears, check that interfaces are up, IP addresses are in the same subnet on each link, OSPF network statements match the interface IPs, all routers use the same area on the link, and no interface is configured as passive. Allow a short time for adjacency formation.

Changing OSPF Cost

To test path selection, increase the cost on one interface that forms part of a route. For example, on R1, enter the following on the R1–R2 interface:

```
R1# configure terminal
R1(config)# interface gigabitEthernet0/0
R1(config-if)# ip ospf cost 100
```

```
R1(config-if)# end
R1# show ip ospf interface gigabitEthernet0/0
R1# show ip route ospf
```

Compare the OSPF routes before and after changing the cost. OSPF prefers the path with the lowest total cost, so traffic may use the alternative route through the third router if that route now has a lower total cost. The exact result depends on all interface costs and the destination being checked. Restore the default by entering no ip ospf cost under the interface.

Results / Observation Table

Test	Command	Observation to record
Interface status	show ip interface brief	Record whether each interface is up/up.
Basic connectivity	ping <neighbor-IP>	Record successful or failed pings.
OSPF neighbors	show ip ospf neighbor	Record neighbor router IDs and states.
OSPF routes	show ip route ospf	Record learned routes and next hops.
Cost change	ip ospf cost 100	Record whether the selected route or metric changes.

Conclusion

OSPF was studied and configured in a three-router triangle topology. Interface addressing and ping tests verify basic connectivity, the neighbor command confirms adjacency formation, and the OSPF routing table shows learned routes. Modifying an interface cost demonstrates how OSPF selects a lowest-cost path. Fill in the observation table using the actual results from your Packet Tracer lab.